

Notes: Heimer & Simsek (JFE, 2019)

Should retail investors' leverage be limited?

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1 Big picture

- **Question.** Does leverage supplied to retail traders improve market quality, or does it mainly finance socially wasteful speculation that enriches intermediaries?
- **Setting.** The paper studies the **U.S. retail foreign-exchange market** around the **2010 CFTC leverage cap**. Starting in October 2010, U.S. traders could use at most **50:1 leverage on major currency pairs** and **20:1 on others**. European retail forex traders remained largely unconstrained and serve as the main control group.
- **Core empirical challenge.** Restricting leverage could reduce speculative losses, but it could also reduce liquidity if high-leverage traders supply information or order flow that helps the market function. The paper therefore has to measure **trading volume, trader returns, brokerage outcomes, and spreads** together.
- **Why this setting is unusually clean.** Retail forex traders are tiny relative to the global FX market, so the leverage cap is unlikely to move aggregate exchange rates. That makes it easier to study the policy's effect on **local intermediation, trader performance, and broker spreads** without simultaneously needing to explain price-level changes in the underlying assets.
- **Main conceptual contribution.** The paper combines reduced-form evidence with a **stylized model of dogmatic-belief traders and competitive brokers**. The welfare lens is **belief-neutral welfare**: even if the planner does not know which traders are “right,” speculation can still be socially harmful because it uses real resources in intermediation and induces excessive risk taking.
- **Main data.** The analysis combines **three data sets**: transaction-level data from a social-networking website linked to retail forex accounts, monthly **CFTC filings** on broker capital and retail-forex obligations, and **interbank quote data** for measuring the broker's execution prices relative to wholesale FX prices.
- **Main empirical design.** The core design is a **difference-in-differences**: compare U.S. traders to European traders before and after the October 2010 cap. The paper also compares **high- versus low-leverage traders** and **forex brokers versus non-forex brokers** regulated by the CFTC.
- **Main trading-volume result.** The actual leverage cap reduces trading volume by about **23%**. For traders predisposed to using high leverage, the reduction is even larger, about **30%**.
- **Main return result.** The policy sharply improves outcomes for the most levered traders. High-leverage U.S. traders improve their relative monthly performance by about **18 percentage points**, which corresponds to mitigating their pre-policy losses by roughly **40%**.
- **Main brokerage result.** Brokerage **excess capital** falls by roughly **25%** after the cap. The decline is strongest for **smaller** brokers and for brokers that previously offered especially high leverage.
- **Main liquidity result.** The policy does **not** significantly change the **relative bid-ask spreads** charged by retail brokers compared with interbank quotes.

- **Key announcement result.** The CFTC's **January 2010 announcement** that it intended to restrict leverage does not meaningfully change trading volume, returns, broker capital, or spreads. The binding change is the **actual constraint**, not the warning.
- **Economic interpretation.** In this market, leverage looks less like a socially useful input into liquidity provision and more like a way to **amplify costly speculation**. Traders lose less, brokers earn less, and liquidity does not worsen.
- **Main takeaway.** The paper concludes that retail leverage in FX was largely facilitating **socially excessive intermediation**. A leverage cap reduces wasteful trading and improves belief-neutral welfare without creating measurable liquidity costs.

2 Data and measurement (institutional context and key objects)

2.1 How the retail forex market works

- Retail forex brokerages are **market makers**. They continuously post bid and ask quotes to clients and are the direct counterparty on trades.
- Brokerages generally do **not** charge separate commissions per trade. Instead, clients pay through the **spread**.
- Traders can also borrow from the brokerage and take positions much larger than their own account balances. That borrowed amount is the source of **leverage**.
- Example: a trader can take a **EUR 100,000** EUR/USD position with only **EUR 20,000** of own capital, which is **5:1 leverage**.
- Clients fund their accounts from domestic bank accounts, but trades are quoted in standard currency-pair notation, such as EUR/USD. Traders do not physically receive the foreign currency; the account records gains and losses.
- By 2010, retail forex turnover was estimated at roughly **USD 125–150 billion per day**, comparable to daily turnover on the full NYSE family of exchanges.

Interpretation. This is a natural market in which to study leverage because leverage is a central product that brokers actively sell to retail clients, not a small margin detail around otherwise unlevered trading.

2.2 Why this market is a good laboratory for policy evaluation

- The market experienced a **discrete regulatory change** in the U.S. in 2010.
- European regulators did **not** impose a similar cap at the same time.

- Many brokerages serve clients from multiple countries, so U.S. and European traders often use the **same type of platform**, trade the **same currency pairs**, and face **overlapping trading hours**.
- Retail forex traders are small relative to the global FX market, so their leverage constraint is unlikely to shift the underlying exchange-rate process itself.

Interpretation. The institutional environment gives the paper both **time-series variation** and a strong **cross-sectional control group**, which is exactly what a clean difference-in-differences design needs.

2.3 The 2010 CFTC regulation and policy timeline

- Before Dodd-Frank, the U.S. retail forex market was only lightly regulated with respect to leverage.
- On **January 20, 2010**, the CFTC publicly proposed a very strict **10:1** leverage limit on all pairs.
- Dodd-Frank became law on **July 21, 2010**, strengthening the CFTC's authority.
- On **September 10, 2010**, the CFTC issued final rules.
- Brokerages had to comply by **October 18, 2010**.
- The final rule capped leverage at **50:1** on major currency pairs and **20:1** on others.
- European regulatory authorities continued to leave leverage mostly to broker discretion, and available leverage almost always exceeded **50:1**.

Interpretation. The paper can separately study the **announcement** and the **implementation**. This is useful because if warnings alone changed behavior, then the effect of the actual cap might be smaller.

2.4 Why traders cannot easily arbitrage across legal regimes

- Brokerages must comply with domestic rules in each country where they operate.
- They therefore segment clients by **country of origin**.
- Verification uses government-issued identification and a link to a **domestic bank account**.

Interpretation. This makes it costly for U.S. traders to simply switch into the “European regime” after regulation, which strengthens the design.

2.5 Transaction-level data from retail forex accounts

- The main micro data come from a social-networking website the paper calls **myForexBook**.

- Registration with the site is free, but it requires an open account on one of the retail forex brokerages linked to the platform.
- By the time of the regulation, around **45 brokerages** were linked; the broader platform covered about **70 retail-specific brokerages**.
- Because myForexBook directly accesses the trading records at each brokerage, even earlier trades can be imported once the trader joins.
- New trades are recorded **automatically** and time-stamped to the **second**, so there is little concern about self-reporting bias.
- The full database contains **8,735 traders** and roughly **5.5 million trades** from early 2009 through April 2012.

The account-level data provide:

- trader location,
- account currency,
- trade direction, currency pair, size, open price, close price, and second of execution,
- daily balances and deposits/withdrawals.

Interpretation. The data are unusually rich for retail FX. They let the authors measure **volume**, **leverage exposure**, and **portfolio returns** rather than just one of these margins.

2.6 Representativeness of the transaction sample

- The authors compare myForexBook traders to a second proprietary data set from one of the world's largest forex brokerages.
- In both data sets, traders who use more leverage have **worse returns**.
- Trading volume in the myForexBook data also covaries strongly with changes in aggregate retail-forex obligations from the CFTC files, with a Pearson correlation of about **41%**.

Interpretation. Even though traders self-select into the social-networking environment, the main leverage-return relationship does not look like an artifact of the sample.

2.7 Sample construction for the main tests

- For the main regulation analysis, the paper keeps only traders who report being in the **United States** or **Europe**.
- It focuses on a symmetric **12-month window** around implementation: **May 2010 to April 2011**.
- This sample contains **15,125 trader-month observations** from **2,672 traders**.
- Of these traders, **1,193** are from the U.S. and **1,479** are from Europe.

- For the announcement analysis, the sample runs from **August 2009 to July 2010** and contains **13,833 trader-month observations**.
- The median trader appears for only **four months**, reflecting both natural attrition and the fact that traders join the social network at different times.

Interpretation. The trimming is designed to minimize contamination from unrelated macro shocks and from slow-moving learning effects in trading behavior.

2.8 Are U.S. and European traders comparable?

- Yes, on observables they look very similar before the policy.
- Experience distributions are close: most traders have **0–1** or **1–3** years of experience.
- About **65%** of both groups say they use **technical trading** strategies.
- The average age is about **37** in both groups.
- U.S. traders average about **365 trades per account**; Europeans average about **307**. Both average roughly **6.6–7.0 trades per trading day**.
- Both groups trade roughly **eight** distinct currency pairs.
- Roughly **16%** of U.S. trades and **19%** of European trades use leverage above **50:1**.
- A probit regression using observables has a very small pseudo- R^2 of about **0.006**, suggesting the treatment/control split is weakly predicted by observed characteristics.
- Pre-policy daily fluctuations are highly correlated across groups: about **0.59** for volume, **0.62** for number of trades, and **0.44** for returns.

Interpretation. This is the backbone of the paper’s identification. The U.S. and European retail traders look similar enough, and move together enough, that Europe is a plausible counterfactual for what would have happened to U.S. traders absent the cap.

2.9 Brokerage-level data from CFTC filings

- The broker-level panel comes from monthly **CFTC Futures Commission Merchants Financial Reports**.
- The CFTC created a new category of registrant, **retail foreign exchange dealers (RFEDs)**, in connection with the 2010 rules.
- Firms serving retail forex customers are required to maintain net capital of **USD 20 million plus 5% of liabilities to retail forex customers above USD 10 million**.
- The panel contains **124 brokerages** around the October 2010 restriction.
- Using post-November 2010 reporting of retail-forex obligations, the paper classifies **24 brokerages** as retail-forex brokers.

- The main broker outcome is **excess capital**, that is, capital in excess of the regulatory minimum.

Interpretation. Excess capital is the paper’s proxy for the size and profitability of the intermediary sector. If the leverage cap shrinks socially excessive intermediation, this should show up here.

2.10 Interbank quote data and the market-liquidity measure

- To measure spreads, the paper merges trader executions with proprietary interbank quote data from **Tick Data**.
- Tick Data records intra-second quotes from roughly **80 banks** active in wholesale FX.
- The authors match each trade to the most competitive interbank price in that second – the highest bid or lowest ask.
- Roughly **two-thirds** of trades can be matched to the **exact second**.

Interpretation. This is what lets the paper move from saying “traders lost money” to saying something sharper about whether **retail execution quality** changed after the policy.

2.11 Main variables

- **Trading volume:** USD value of positions opened by trader i in month t .
- **Number of trades:** count of trades opened by trader i in month t .
- **Monthly portfolio return:** change in account balance net of deposits/withdrawals.
- **High-leverage trader:** indicator for being above the sample median of pre-policy average leverage use.
- **Excess capital:** broker capital above the regulatory requirement.
- **Relative bid-ask spread:** the retail broker’s execution-price disadvantage or advantage relative to the interbank market, normalized by price.

Interpretation. The empirical strategy maps almost perfectly to the theory: trader outcomes, intermediary outcomes, and liquidity outcomes are all observed directly.

3 Models and empirical specifications (all equations + interpretation)

3.1 Stylized theory of leverage, speculation, and intermediation

The paper’s model is deliberately stripped down but economically sharp:

- There is a single traded asset.

- Traders have **heterogeneous and dogmatic beliefs**. They believe their own view is correct and do not learn from prices.
- Some beliefs may contain useful information, but many are simply **noise** or behavioral distortions such as overconfidence.
- Traders choose speculative positions, often with leverage.
- A competitive brokerage sector intermediates these positions and incurs:
 - **technological intermediation costs**, like infrastructure and labor,
 - **informational costs**, because traders may have private information.

The main economic implications are:

- More trading volume means traders pay more intermediation costs through spreads, so their **expected net return falls**.
- More trading volume increases broker revenue and the size of the brokerage sector.
- Belief-neutral social welfare falls with **trading volume** and **portfolio risk**.
- A leverage cap reduces volume and risk, which raises traders' expected net returns and lowers broker revenue.
- The effect on bid-ask spreads is **ambiguous**: spreads rise only if cutting high-leverage trades makes the average order flow more informative.

Interpretation. The policy is predicted to improve welfare through a simple channel: less leverage means fewer wasteful speculative round-trips that absorb real intermediary resources. But the data still have to show whether the policy also harms liquidity.

3.2 Baseline difference-in-differences for trading volume

The main trader-level regression is

$$\log(\text{TradingVolume})_{it} = \gamma_i + \gamma_t + \beta_1 (US_i \times \text{PostConstraint}_t) + \varepsilon_{it}. \quad (1)$$

- i indexes traders and t indexes months.
- $US_i = 1$ for U.S. traders and 0 for European traders.
- $\text{PostConstraint}_t = 1$ after the October 2010 implementation and 0 before.
- γ_i are trader fixed effects.
- γ_t are calendar-month fixed effects.

Interpretation. β_1 is the average treatment effect of the leverage cap on the volume of U.S. trading relative to the European control group.

3.3 Dynamic event-study around implementation

To examine timing and pre-trends, the paper estimates

$$\log(\text{TradingVolume})_{it} = \gamma_i + \gamma_t + \sum_{k=T-\ell}^{T+\ell} \beta_k (US_i \times I_{T+k=t}) + \varepsilon_{it}. \quad (2)$$

Interpretation. The coefficients β_k trace out the month-by-month path of the treatment effect around the implementation date T . Flat pre-coefficients support the parallel-trends assumption; a sharp post-drop supports a causal interpretation.

3.4 High-leverage heterogeneity regression

To test whether the policy matters most where it should, the paper estimates

$$\log(\text{TradingVolume})_{it} = \gamma_i + \gamma_t + \beta_1 (\text{HighLev}_i \times \text{PostConstraint}_t) + \beta_2 \text{Controls}_{it} + \varepsilon_{it}. \quad (3)$$

Here $\text{HighLev}_i = 1$ if trader i was above the median of pre-policy average leverage use.

Interpretation. If leverage is the real mechanism, the effect should be stronger for the traders who were most levered to begin with.

3.5 Announcement regression

To separate warnings from actual legal restrictions, the paper estimates

$$\log(\text{TradingVolume})_{it} = \gamma_i + \gamma_t + \beta_1 (US_i \times \text{PostAnnouncement}_t) + \varepsilon_{it}, \quad (4)$$

where $\text{PostAnnouncement}_t = 1$ after the CFTC's January 2010 proposal.

Interpretation. This asks whether disclosure and warnings alone are enough to deter speculative trading, or whether only an enforceable leverage cap matters.

3.6 Portfolio-sort design for trader returns

The paper sorts traders each month into **five portfolios** using the trader's average leverage use in month $t - 1$. Let

$$HML_t = R_t^{Q5} - R_t^{Q1}, \quad (5)$$

where $Q5$ is the highest-leverage quintile and $Q1$ is the lowest-leverage quintile.

The paper's natural treatment effect is the double difference

$$\widehat{LATE} = (\bar{R}_{US,post} - \bar{R}_{US,pre}) - (\bar{R}_{EU,post} - \bar{R}_{EU,pre}). \quad (6)$$

It also computes one-factor alphas by regressing high-minus-low returns on a carry-trade factor:

$$HML_t = \alpha + \beta Carry_t + u_t. \quad (7)$$

Interpretation. The return analysis is not just a before/after mean comparison. It is designed to show that the policy disproportionately helps the traders most exposed to leverage, and that the result is not merely compensation for systematic FX risk.

3.7 Broker-level regression for excess capital

The main brokerage specification is

$$\log(\text{ExcessCapital})_{bt} = \gamma_b + \gamma_t + \beta_1 (FXBroker_b \times PostConstraint_t) + Controls_{bt} + \varepsilon_{bt}, \quad (8)$$

where b indexes brokerages and $FXBroker_b = 1$ if the firm has retail-forex obligations.

The event-study version is

$$\log(\text{ExcessCapital})_{bt} = \gamma_b + \gamma_t + \sum_{j=-6}^7 \beta_j (FXBroker_b \times I_{T+j=t}) + \varepsilon_{bt}. \quad (9)$$

Interpretation. These regressions ask whether the leverage cap shrinks the intermediary sector that profits from retail speculation.

3.8 Relative bid-ask spread construction

For a long trade opened at t and closed at t' , the paper defines

$$S_t = \frac{P_t^{broker} - P_t^{interbank,ask}}{P_t^{interbank,ask}} + \frac{P_{t'}^{interbank,bid} - P_{t'}^{broker}}{P_{t'}^{interbank,bid}}. \quad (10)$$

For a short trade,

$$S_t = \frac{P_t^{interbank,bid} - P_t^{broker}}{P_t^{interbank,bid}} + \frac{P_{t'}^{broker} - P_{t'}^{interbank,ask}}{P_{t'}^{interbank,ask}}. \quad (11)$$

Interpretation. S_t is the all-in execution disadvantage of trading through the retail broker rather than at interbank quotes, expressed as a fraction of price.

3.9 Difference-in-differences for spreads

The spread regression is

$$\text{RelativeSpread}_{jit} = \gamma_i + \gamma_t + \beta_1 (US_i \times PostConstraint_t) + \beta_2 Controls_{jit} + \varepsilon_{jit}, \quad (12)$$

where j indexes trades.

- The regressions are at the **trade level**.
- They include **day fixed effects**.
- Some specifications use only trades matched **exactly to the second**; others use all matches while controlling for the log distance of the match.

Interpretation. This is the paper’s direct test of whether the policy worsened local market liquidity.

3.10 Estimation choices and inference

- Trader-month regressions double-cluster standard errors by **trader and month**.
- Broker regressions double-cluster by **broker and month**.
- Spread regressions double-cluster by **trader and day**.
- Some broker regressions use **entropy balancing** to match the size distribution of control firms to FX brokers.
- Spread regressions winsorize the dependent variable at the **1%** level.

Interpretation. The estimation choices are sensible for the panel structure and show that the paper takes inference seriously rather than relying on one fragile specification.

4 Identification and empirical design (what makes the estimates credible)

4.1 Why the U.S.-Europe comparison is the paper’s cleanest design

- U.S. and European traders often trade the same pairs, especially EUR/USD.
- Their trading hours overlap, so they face similar market conditions.
- They look similar on age, experience, trading style, and baseline trading intensity.
- The brokerage architecture is similar across the two groups.
- Regulatory arbitrage is hard because country verification depends on identity documentation and domestic bank links.

Why this matters. The treatment is not “being American” in any general sense; it is exposure to a leverage cap that Europeans did not face at the same time.

4.2 Parallel trends are explicitly tested

- Daily fluctuations in trading volume, number of trades, and returns are highly correlated across U.S. and European traders before the cap.
- The event-study coefficients for trading volume are close to **zero** before October 2010.
- Similar common-trend evidence is shown for brokerage capital.
- Trade-level return tests in the Internet Appendix also satisfy common-trend checks.

Why this matters. Difference-in-differences is only credible if treatment and control would have moved together absent the policy. The paper spends real effort documenting that this assumption is plausible.

4.3 The announcement result is an important credibility check

- The January 2010 CFTC proposal drew attention – Google searches for “forex leverage” spiked.
- Yet the announcement does not significantly reduce trading volume, improve returns, reduce brokerage capital, or change spreads.

Why this matters. This makes it less likely that the paper is just picking up a broad change in sentiment about forex trading or a generic time trend in the U.S. market. The timing matches the **binding legal cap**, not discussion alone.

4.4 Treatment intensity lines up with the mechanism

- High-leverage traders reduce trading volume more than low-leverage traders.
- Smaller brokers and brokers that previously offered especially high leverage experience the largest capital declines.

Why this matters. When the treatment bites hardest exactly where theory says it should, the causal story becomes much more persuasive.

4.5 Data-quality concerns are addressed directly

- The transaction data are automatically pulled from broker records, which limits self-reporting or recall bias.
- The authors validate myForexBook traders against another large-broker data set.
- Aggregate myForexBook trading volume tracks independent CFTC measures of retail-forex obligations.

Why this matters. The paper’s strongest results are only useful if the account-level data are not some peculiar niche corner of the market. The validation exercises make that concern less severe.

4.6 Why the broker-capital result is interpreted as a revenue effect

A decline in excess capital could in principle come from lower expected future profitability and worse access to external financing, not necessarily from lower current intermediation revenues. The paper argues against that story with three facts:

- the **announcement** does not reduce broker capital,
- the decline is stronger for **smaller brokers**, which are precisely the firms that are most vulnerable when trading volume falls,
- the policy reduces **new client entry** more than it reduces exits, lowering the customer base by about **1% per broker-month**.

Why this matters. The broker-capital results are not just a valuation story. They look like a real contraction in intermediation activity.

4.7 Why the paper's welfare language is more careful than a standard rational benchmark

- Traders may sincerely believe their trades are good ideas.
- The planner does not need to know whose beliefs are right to say that resources spent intermediating speculation are socially costly.
- The paper therefore uses **belief-neutral welfare**, which strips out speculative transfers between traders and focuses on intermediation costs and portfolio risk.

Why this matters. This is the paper's answer to the usual objection that "maybe these traders simply disagree and some of them are right." Even under disagreement, excessive intermediation can be wasteful.

4.8 What the paper cannot fully distinguish

- It cannot observe all motives for trading. Some traders may hedge background risks or simply enjoy trading.
- The gross-return calculations depend on assumed trading-cost levels of **2 to 5 pips**, so they are back-of-the-envelope rather than exact.
- The stylized model is not a full structural model of market making, entry, or retail demand.

Bottom line. The paper convincingly identifies a **reduced-form causal effect** of the leverage cap in retail forex. Its stronger welfare claims are guided by the model and are plausible, but they still rely on the paper's interpretation of why retail traders are in this market.

5 Tables and figures

5.1 Table 1: U.S. and European traders look comparable

Read:

- U.S. and European traders have nearly identical experience distributions, similar strategy choices, and the same average age of about **37**.
- Trading intensity is also close: roughly **365** trades per U.S. account versus **307** per European account in the pre-policy window.
- The pre-policy fraction of trades above **50:1** leverage is substantial in both groups: **16%** for U.S. traders and **19%** for Europeans.
- Aggregate daily changes in activity move together strongly: volume correlation **0.59**, number of trades **0.62**, returns **0.44**.

Economic message. The paper's control group is not just convenient – it is genuinely similar to the treated group on the margins that matter most for this question.

Table 1
A comparison balancing test.
This table compares traders in the myforexbook data set from the United States to traders from Europe. Panel A includes a comparison of means. Panel B presents Pearson's correlation coefficients from daily fluctuations in aggregate trading activity between U.S. and European traders (weekends excluded). Daily fluctuations equal $\log(\hat{y}_t/\hat{y}_{t-1})$ where $\hat{y}$ is the average by day t , and i indexes U.S. versus European traders. The sample window for these tests is the period immediately preceding the CFTC leverage-constraint – May 1 to October 17, 2010.

Panel A: Difference in means between U.S. and European traders			
	US	EUR	t-stat*
Personal characteristics			
Trading experience (years)			
0-1	0.29	0.31	1.17
1-3	0.45	0.44	0.40
3-5	0.10	0.10	0.25
5+	0.15	0.14	0.76
Trading approach			
Technical	0.65	0.66	0.66
Fundamentals	0.046	0.053	0.78
News	0.024	0.020	0.70
Not specific	0.23	0.22	0.62
Momentum	0.053	0.048	0.57
Age	37.4	37.4	0.09
Trading characteristics			
Trades per account	364.6	307.3	0.68
Trades per account/day, if ≥ 1 trade in day	6.59	7.03	0.60
Distinct currency pairs traded at least once	8.46	8.17	1.15
Fraction trades w/ leverage > 50:1	0.16	0.19	2.67
Number of traders	1193	1479	
* test of equality of means between US and EUR			
Panel B: Correlation of aggregate daily trading for U.S. versus Europeans			
			Corr. coef.
Trading volume (USD)			0.59
Number of trades			0.62
Returns			0.44

Table 2
Summary statistics from traders' accounts.

This table presents summary statistics from the myforexbook account-level database trimmed according to the criteria described in Section 3. The sample includes trades executed by U.S. and European retail forex traders. US trader equals one if the trader is located in the U.S. and equal to zero if located in Europe. Post constraint equals one if the month comes after October 2010, the month in which brokerages needed to comply with CFTC regulations limiting the leverage available to US retail forex traders at 50:1, zero otherwise. Post announcement equals one if the trade was opened after the CFTC's announcement in the Federal Register in January 2010 of their intent to restrict traders' leverage to 10:1, zero otherwise. High-leverage trader equals one if the trader's average leverage use prior to the leverage-constraint is above the median of all traders in the sample. Monthly portfolio returns are calculated using the account's balances at the beginning and end of the month, excluding deposits.

Panel A: Sample window around leverage-constraint (May 2010–April 2011)									
Variable	Mean	Std dev	5%	10%	25%	50%	75%	90%	95%
Log trading volume (USD)	11.7	2.65	7.14	8.29	10.00	11.8	13.5	14.8	15.7
Log number of trades	3.31	1.62	0	1.10	2.20	3.43	4.44	5.28	5.81
Portfolio return	-0.19	0.90	-1.42	-1.00	-0.34	-0.0092	0.020	0.23	0.70
US trader (= 1)	0.46								
Post constraint (= 1)	0.54								
High-leverage trader (= 1)	0.55								
Trader-month observations	15,125								
Panel B: Sample window around regulation announcement (August 2009–July 2010)									
Log trading volume (USD)	11.9	2.46	7.67	8.85	10.3	12.1	13.6	15.0	15.8
Log number of trades	3.31	1.55	0.69	1.10	2.30	3.47	4.42	5.16	5.62
Portfolio return	-0.31	1.05	-1.98	-1.22	-0.74	-0.046	0.013	0.26	0.89
US trader (= 1)	0.47								
Post announcement (= 1)	0.69								
Trader-month observations	13,833								

5.2 Tables 2 and 3 plus Figure 1: the leverage cap sharply reduces trading volume

Read:

- In the main trader fixed-effect specification, the coefficient on $US \times PostConstraint$ is **-0.265**, which translates to about a **23%** decline in trading volume.
- With brokerage-month fixed effects, the coefficient is **-0.200**, still economically large and statistically significant.
- On the number of trades, the effect is smaller but still negative: around **-0.108**.
- For high-leverage traders, the volume effect is much larger: around **-0.365** to **-0.340**, or roughly a **30%** decline.

- The announcement coefficients are small and statistically insignificant.
- Figure 1 shows flat pre-trends and then a sharp break right after implementation.

Table 3
Leverage constraints and trading volume.
This table uses account level data from the myforexbook data set collapsed to the level of trader-month. It reports ordinary least squares estimates of the regression

$$\log(\text{volume})_{it} = \gamma_i + \gamma_t + \beta_1 \text{US trader}_i \times \text{post constraint}_t + \varepsilon_{it}$$

where i is a trader and t is a month. Trading volume is the USD value of the assets traded. US trader equals one if the trader is from the U.S. and equal to zero if they are from Europe. Post constraint equals one in months starting in November 2010, and zero otherwise. The sample period in Panel A and Panel B is May 2010 to April 2011. The sample period in Panel C is August 2009 to July 2010. Standard errors are double-clustered by trader and month, and *, **, and *** denote significance at the $p < 0.1$, $p < 0.05$ and $p < 0.01$ levels, respectively.

Panel A: The October 2010 leverage constraint and trading volume				
Dep var:	Log trading volume		Log number of trades	
	(1a)	(2a)	(3a)	(4a)
US trader (=1) × Post constraint (=1)	-0.265*** (0.094)	-0.200** (0.097)	-0.108* (0.062)	-0.0685 (0.065)
Trader FE	x	x	x	x
Month FE	x	x	x	x
Brokerage-month FE	x	x	x	x
Trader-month observations	15,125	15,016	15,156	15,047
R ²	0.72	0.73	0.61	0.61
Panel B: Alternative treatment and control groups –high- vs. low-leverage traders				
Dep var:	Log trading volume		Log number of trades	
	(1b)	(2b)	(3b)	(4b)
High leverage trader (=1) × Post constraint (=1)	-0.365*** (0.094)	-0.340*** (0.096)	-0.155** (0.062)	-0.154** (0.063)
Trader FE	x	x	x	x
Month FE	x	x	x	x
Brokerage-month FE	x	x	x	x
Trader-month observations	15,125	15,016	15,156	15,047
R ²	0.72	0.73	0.61	0.61
Panel C: The January 2010 announcement of leverage regulation				
Dep var:	Log trading volume		Log number of trades	
	(1c)	(2c)	(3c)	(4c)
US trader (=1) × Post announcement (=1)	0.148 (0.10)	0.153 (0.12)	0.0433 (0.075)	0.0998 (0.080)
Trader FE	x	x	x	x
Month FE	x	x	x	x
Brokerage-month FE	x	x	x	x
Trader-month observations	13,833	13,741	13,861	13,767
R ²	0.70	0.71	0.55	0.57

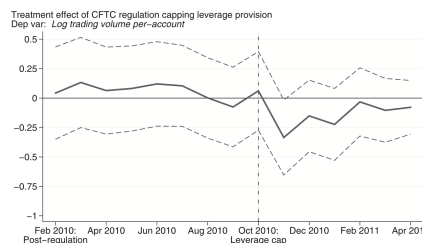


Fig. 1. Impulse response of treatment effect on trading volume. This table reports ordinary least squares estimates of the following regression: $\log(\text{trading volume})_{it} = \gamma_i + \gamma_t + \sum_{j=1}^{T-1} \beta_j \text{US trader}_i \times I_{t-j,10,10} + \varepsilon_{it}$, where i is a trader and t is a month. The dependent variable is $\log(\text{trading volume})$, which is the natural logarithm of total monthly trading volume in USD. US trader equals one if the trader is located in the U.S. and equal to zero if located in Europe. T is the date of the regulation, i.e. October 2010. $I_{t-j,10,10}$ is an indicator variable for months surrounding the regulation. Therefore, β_j for $j = 1, \dots, T-1$ is the sequence of treatment effects, and hence maps out the impulse response. Standard errors are double-clustered by day and trader, and the dashed lines are 95% confidence intervals around the point estimate of β_j .

Economic message. The leverage cap did exactly what it was intended to do on the activity margin: it **shrunk speculative trading**, especially among the traders most likely to rely on leverage.

5.3 Figure 2 and Table 4: leverage is strongly associated with poor retail performance

Read:

- Before the cap, the highest-leverage U.S. portfolio loses about **44% per month**; the lowest-leverage portfolio loses only about **3%**.
- The average U.S. portfolio improves from roughly **-17.4%** pre-policy to **-9.5%** post-policy.
- The high-minus-low return for U.S. traders improves from about **-41.2 percentage points** to about **-17.5 percentage points**.
- After differencing against Europe, the local treatment effect on the high-minus-low return is about **18 percentage points**.
- The one-factor alpha version tells essentially the same story: the high-minus-low alpha improves by about **19 percentage points** and is statistically significant.
- The announcement-period return tables show little comparable improvement.

Economic message. Leverage is not just correlated with bigger gains and bigger losses in a symmetric way. In this market, the most levered traders are persistently the **worst performers**, and the cap substantially protects them from those losses.

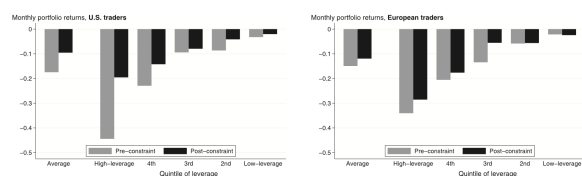


Fig. 3. Leverage constraints and portfolio returns. This table reports average portfolio returns of traders in the myforebook database. Monthly returns are calculated using the account's balances at the beginning and end of the month, excluding deposits. We sort traders monthly by the average amount of leverage the trader uses in $t-1$.

Table 4
Leverage constraints and trader's portfolio returns. This table reports portfolio returns of traders in the myforebook database. Monthly returns are calculated using the account's balances at the beginning and end of the month, excluding deposits. We sort traders monthly by the average amount of leverage the trader uses in $t-1$. We calculate one-factor alphas by regressing the return difference between the high- and low-leverage portfolios (high minus low return) on the carry trade factor described in Brunner et al. (2015). To calculate statistical differences between pre- and post-constraint portfolios, we use seemingly unrelated regressions (joint equations regressions) to conduct χ^2 tests on the regression alphas; the table presents p-values for the null hypothesis that the difference in alpha is equal to zero. Stars *, **, and *** denote significance levels $p < 0.10$, $p < 0.05$, and $p < 0.01$, respectively.

Panel A: Sample window around leverage constraint (May 2010–April 2011)											
US traders						European traders					
	Pre-constraint	Post-constraint	Post minus pre	t-stat		Pre-constraint	Post-constraint	Post minus pre	t-stat		Local auc. treatment effect ($U_{H_{post}} - U_{H_{pre}}$) minus ($U_{L_{post}} - U_{L_{pre}}$)
Sample average	-0.174	-0.095	0.079**	[4.46]		-0.140	-0.110	0.030	[1.53]		0.05
Leverage quintile											
High	-0.444	-0.195	0.249**	[4.02]		-0.340	-0.285	0.055	[0.87]		0.06
1	-0.229	-0.142	0.087**	[2.38]		-0.205	-0.170	0.035	[0.44]		0.04
2	-0.204	-0.079	0.124	[0.44]		-0.154	-0.055	0.099**	[2.38]		-0.06
3	-0.286	-0.041	0.245**	[3.64]		-0.028	-0.016	0.012	[0.05]		0.04
Low	-0.032	-0.020	0.012	[0.40]		-0.024	-0.024	-0.003	[0.09]		0.01
High minus low	-0.412**	-0.175**	0.237**	[4.02]		-0.316**	-0.261**	0.055	[0.87]		0.16
1-factor alpha	[5.75]	[4.07]	0.237**	[5.84]		[4.35]		0.043			0.19**
	-0.404**	-0.190**	0.214**	[3.64]		-0.280**	-0.250**	0.030	[1.53]		χ^2 p-val = 0.62
	[30.02]	[1.64]		[3.70]		[3.70]					χ^2 p-val = 0.01

Panel B: Sample window around regulation announcement (August 2009–July 2010)											
US traders						European traders					
	Pre-announcement	Post-announcement	Post minus pre	t-stat		Pre-announcement	Post-announcement	Post minus pre	t-stat		Local auc. treatment effect ($U_{H_{post}} - U_{H_{pre}}$) minus ($U_{L_{post}} - U_{L_{pre}}$)
Sample average	-0.145	-0.109	-0.023	[0.83]		-0.193	-0.213	-0.020	[0.66]		0.00
Leverage quintile											
High	-0.392	-0.453	-0.066	[0.66]		-0.420	-0.518	-0.098	[0.07]		0.03
1	-0.224	-0.302	-0.078	[0.90]		-0.305	-0.265	0.040	[1.04]		-0.06
2	-0.018	-0.078	-0.060	[1.90]		-0.153	-0.180	-0.027	[0.52]		-0.03
3	-0.044	-0.177	-0.133	[0.83]		-0.069	-0.104	-0.035	[0.76]		-0.01
Low	-0.027	-0.050	-0.023	[0.21]		0.032	-0.051	-0.083	[1.41]		0.05
High minus low	-0.365**	-0.397**	-0.032**	[5.07]		-0.452**	-0.467**	-0.015	[0.76]		-0.02
1-factor alpha	[3.05]	[3.80]	-0.007	[0.40]		[5.07]		0.203			-0.27*
	-0.340**	-0.414**	-0.074**	[4.12]		-0.462**	-0.509**	-0.047**	[1.00]		χ^2 p-val = 0.49
	[4.01]	[4.12]		[5.00]		[5.00]					χ^2 p-val = 0.05

5.4 Back-of-the-envelope gross returns: most of the damage comes from trading costs

Read:

- Using historical spreads of **2 pips**, the authors estimate that about **60%** of high-leverage traders' losses are due to transaction costs.
- With assumed costs of **3 to 4 pips**, the gross-return gap between high- and low-leverage traders becomes small and statistically insignificant.
- With **5 pips**, the high-leverage group can even have better gross performance than the low-leverage group.

Economic message. The paper's strongest interpretation is that leverage magnifies **fee-paying speculation**. The traders are not necessarily catastrophically bad at directional forecasting on a gross basis; they are mostly harmed by trading **too much**, at **too much leverage**, while paying spreads over and over.

5.5 Tables 5 and 6 plus Figures 3 and 4: the brokerage sector shrinks after the cap

Read:

- In broker-level regressions, the treatment coefficient is between about **-0.276** and **-0.313**, implying about a **25%** decline in excess capital.
- The effect is much larger for brokers below the median size: around **-0.495** to **-0.548**.
- For above-median brokers, the coefficient is near **-0.10** and insignificant.
- Figure 3 shows no meaningful pre-trend and then a decline beginning around the implementation window.
- Figure 4 shows substantial consolidation in the broker market after the rules: many retail-FX brokers disappear, and only a few large firms survive.

Economic message. The policy did not merely hurt traders' ability to speculate. It also shrank the intermediary sector that was making money from that speculation, which is exactly what the

Table 5
Summary statistics on retail forex brokerages. The data comes from monthly reports to the CFTC. The CFTC requires futures commission merchants and retail foreign exchange dealers to file monthly financial reports with the CFTC's Division of Swap Dealer and Intermediary Oversight. The CFTC collects data on brokerage excess capital and net capital requirements for the entire sample. They begin collecting data on retail forex obligations in November 2010. Net capital requirement is set as a minimum of \$1,000,000 or some greater amount as determined under CFTC Regulation 1.17(a). FX broker equals one if the broker has retail forex obligations greater than zero post-November 2010, and zero otherwise. Post constraint equals one starting in November, 2010, the date by which brokerages needed to comply with CFTC regulation limiting the leverage available to US retail forex traders at 50:1, zero otherwise. Post announcement equals one starting in February 2010, following the CFTC's announcement in the Federal Register of their intent to restrict traders' leverage to 10:1, zero otherwise.

Panel A: Sample window around leverage constraint (May 2010–April 2011)										
Variable	Broker-month obs.	Mean	Std dev	5%	10%	25%	50%	75%	90%	95%
Log excess net capital	1505	174	2.88	13.2	13.6	15.1	17.3	19.5	21.3	22.6
Log net capital requirement	1505	161	2.27	13.8	13.8	13.8	16.0	17.5	19.8	20.6
FX broker (=1)	1505	0.16								
Post constraint (=1)	1505	0.50								
Log retail forex obligations if FX broker = 1	117	16.5	1.79	14.1	14.2	15.0	16.9	17.5	18.8	18.8
Panel B: Sample window around regulation announcement (August 2009–July 2010)										
Variable	Broker-month obs.	Mean	Std dev	5%	10%	25%	50%	75%	90%	95%
Log excess net capital	1571	172	2.98	12.7	13.3	14.9	17.0	19.4	21.2	22.6
Log net capital requirement	1573	15.8	2.37	13.1	13.1	13.8	15.6	17.2	19.2	20.5
FX broker (=1)	1573	0.14								
Post announcement (=1)	1573	0.49								
Log retail forex obligations	No data prior to November 2010									

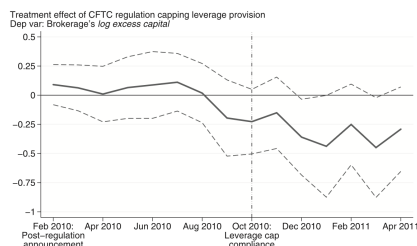


Fig. 3. Impulse response of treatment effect on brokerage excess capital. This table reports ordinary least squares estimates of the following regression: $\log(\text{excess capital})_{it} = \gamma_0 + \gamma_1 + \sum_{j=1}^T \beta_j \text{FX broker}_{it} \times \text{post constraint}_{it} + \epsilon_{it}$, where b is a broker and t is a month. The dependent variable is the logarithm of the brokerages' excess capital, recorded by the CFTC. FX broker equals one if the brokerage has any retail forex obligations after they were required to report these obligations starting in November 2010. T is the date of the regulation, i.e., October 18, 2010. $\text{post constraint}_{it}$ is an indicator variable for months surrounding the regulation. Therefore, β_j for $j = \{-T, \dots, T\}$ is the sequence of treatment effects, and hence maps out the impulse response. Standard errors are double-clustered by month and broker, and the dashed lines are 95% confidence intervals around the point estimate of β_j .

Table 6
The effect of the leverage-constraint and announcement on brokerage excess capital. This table reports OLS estimates of the regression

$$\log(\text{excess capital})_{it} = \gamma_0 + \gamma_1 + \beta_1 \text{FX broker}_{it} \times \text{post constraint}_{it} + \epsilon_{it}$$

where b is a broker and t is a month. The data come from monthly CFTC Futures Commission Merchants Financial Reports. Excess capital is the capital in excess of the regulatory requirement, for each brokerage in the CFTC data set. FX broker_{it} equals one if the brokerage has any retail forex obligations after they were required to report these obligations starting in November 2010, and zero otherwise. $\text{Post constraint}_{it}$ equals one in months starting in November 2010, and zero otherwise. $\text{Post announcement}_{it}$ equals one in months starting in February 2010, and zero otherwise. Using the non-parametric, entropy balancing procedure designed by Hainmueller (2012), some regressions weight observations so that the control group's mean, standard deviation, and skewness match that of the set of the FX brokers in the pre-constraint period. The sample period is May 2010 to April 2011 in Panels A and B, and is August 2009 to July 2010 in Panel C. Standard errors are double-clustered by broker and month, and *, **, and *** denote significance at the $p < 0.1$, $p < 0.05$, and $p < 0.01$ levels, respectively.

Panel A: The October 2010 leverage constraint's effect on brokerage capital				
Dep var: Log excess capital	(1a)	(2a)	(3a)	(4a)
FX broker (=1) × Post constraint (=1)	-0.276** (0.12)	-0.289** (0.14)	-0.312** (0.13)	-0.313** (0.13)
Log net capital requirement		-0.207 (0.17)		-0.00912 (0.19)
Brokerage FE	x	x	x	x
Month FE	x	x	x	x
Sample balanced by broker size	No	No	Yes	Yes
Brokerage-month observations	1503	1503	1456	1456
R ²	0.59	0.59	0.57	0.57
Panel B: The leverage constraint, brokerage capital sorted by broker size				
Dep var: Log excess capital	Sample: Broker size ...		Above median	
	(1b)	(2b)	(3b)	(4b)
FX broker (=1) × Post constraint (=1)	-0.495** (0.23)	-0.548* (0.29)	-0.300 (0.092)	-0.300 (0.093)
Log net capital requirement		-0.367 (0.27)		-0.00250 (0.057)
Brokerage FE	x	x	x	x
Month FE	x	x	x	x
Brokerage-month observations	635	635	821	821
R ²	0.86	0.88	0.98	0.98
Panel C: The January 2010 announcement of leverage regulation				
Dep var: Log excess capital	(1c)	(2c)	(3c)	(4c)
FX broker (=1) × Post announcement (=1)	-0.0542 (0.11)	-0.0456 (0.11)	-0.0418 (0.12)	-0.0463 (0.11)
Log net capital requirement		0.0284 (0.081)		-0.0146 (0.086)
Brokerage FE	x	x	x	x
Month FE	x	x	x	x
Sample balanced by broker size	No	No	Yes	Yes
Brokerage-month observations	1570	1570	1542	1542
R ²	0.98	0.98	0.98	0.98

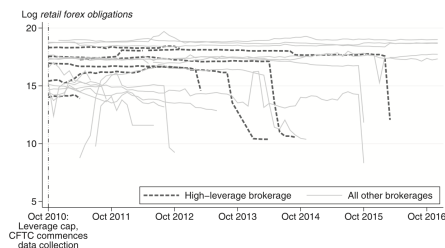


Fig. 4. Brokerages' retail forex obligations. This figure presents each brokerage's retail foreign exchange obligations. Obligations are equal to the total amount of funds at a brokerage that would be obtained by combining all money, securities, and property deposited by a retail forex customer into a retail forex account or accounts, adjusted for the realized and unrealized net profit or loss. The CFTC began requiring this reporting in November 2010. We manually classify brokerages as high-leverage if they were advertised as offering traders at least 400:1 leverage per-trade around the time of the November 2010 CFTC regulation. To generate this classification, we searched Internet archives. Most of the brokerages were listed on the website www.100forexbrokers.com, which provides a directory of forex brokerages and their characteristics.

theory predicts if retail leverage had been generating rents for brokers.

5.6 Tables 7 and 8: no evidence of worse local liquidity

Table 7

Summary statistics on relative bid-ask spreads.

This table presents summary statistics from the myforexbook account-level database trimmed according to the criteria described in Section 3. The sample includes trades executed by U.S. and European retail forex traders. *US trader* equals one if the trade is executed by a trader located in the U.S. and equal to zero if located in Europe. *Post constraint* equals one if the trade was opened after October 18, 2010, the date by which brokerages needed to comply with CFTC regulations limiting the leverage available to U.S. retail forex traders at 50:1, zero otherwise. *Post announcement* equals one if the trade was opened after the CFTC's announcement in the Federal Register on January 13, 2010 of their intent to restrict traders' leverage to 10:1, zero otherwise. The trade's relative bid-ask spread is defined as: relative bid-ask spread_{*it*} = $\left(\frac{p_{it}^{ask} - p_{it}^{bid}}{p_{it}^{ask} + p_{it}^{bid}} \right)$ if the transaction is long, and relative bid-ask spread_{*it*} = $\left(\frac{p_{it}^{bid} - p_{it}^{ask}}{p_{it}^{ask} + p_{it}^{bid}} \right)$ if the transaction is short, where *j* is a trader, *i* is a trade, and *t* is the second the trade is opened. To calculate relative bid-ask spread, we match traders' transactions in the myforexbook data with the closest (in time) quoted price in Tick Data - a data set of interbank foreign exchange quotes.

Panel A: Sample window around leverage constraint (May 1, 2010–April 30, 2011)								
Variable	Mean	Std dev	5%	10%	25%	50%	75%	95%
Relative bid-ask spreads	-0.00035	0.31	-0.49	-0.32	-0.14	-0.022	0.12	0.36
US trader (= 1)	0.47							0.57
post constraint (= 1)	0.47							
Number of trades	1,444,159							

Panel B: Sample window around regulation announcement (August 1, 2009–July 31, 2010)								
Variable	Mean	Std dev	5%	10%	25%	50%	75%	95%
Relative bid-ask spreads	-0.0021	0.31	-0.49	-0.32	-0.13	-0.021	0.11	0.36
US trader (= 1)	0.48							0.57
Post announcement (= 1)	0.69							
Number of trades	1,140,274							

Table 8

The effect of the leverage constraint on brokers' relative bid-ask spreads.

This table reports OLS estimates of the following regression:

$$\text{relative bid-ask spread}_{it} = \gamma_1 + \gamma_2 \text{US trader}_i + \gamma_3 \text{post constraint}_i + \gamma_4 \text{trade}_{it} + \epsilon_{it}$$

where *i* is a trader, *j* is a trade, and *t* is a day (trades are recorded by the second). The dependent variable is relative bid-ask spread_{*it*} = $\left(\frac{p_{it}^{ask} - p_{it}^{bid}}{p_{it}^{ask} + p_{it}^{bid}} \right)$ if the transaction is long, and relative bid-ask spread_{*it*} = $\left(\frac{p_{it}^{bid} - p_{it}^{ask}}{p_{it}^{ask} + p_{it}^{bid}} \right)$ if the transaction is short. US trader equals one if the trade is executed by a trader located in the U.S. and equal to zero if located in Europe. To calculate relative bid-ask spread, we match traders' transactions in the myforexbook data with the closest (in time) quoted price in Tick Data - a data set of inter-bank foreign exchange quotes. Columns 1 and 2 include trades that we can match perfectly to the second, while Columns 3 and 4 include all trades that we can match and the regression controls for the match distance (in seconds). *Post constraint* equals one if the trade was opened after October 18, 2010, the date by which brokerages needed to comply with CFTC regulation limiting the leverage available to U.S. retail forex traders at 50:1, zero otherwise. *Post announcement* equals one if the trade was opened after the CFTC's announcement in the Federal Register on January 13, 2010 of their intent to restrict trader leverage to 10:1, zero otherwise. In Panel A, the sample period is from May 1, 2010 to April 30, 2011. In Panel B, the sample period is from August 1, 2009 to July 31, 2010. Standard errors are double-clustered by day and trader, and *, **, and *** denote significance levels *p* < 0.10, *p* < 0.05, and *p* < 0.01, respectively.

Panel A: The October 2010 leverage constraint and relative bid-ask spreads				
Dep var: Relative bid-ask spreads	(1a)	(2a)	(3a)	(4a)
US trader (=1) × Post constraint (=1)	-0.0105 (0.0080)	0.00559 (0.053)	-0.0138 (0.0085)	0.00145 (5.10)
US trader (=1) × ...				
... Ind[May 2010–Jul 2010]		0.0256 (0.17)		0.0263 (2.71)
... Ind[Aug 2010–Sep 2010]		0.00851 (1.24)		0.013 (0.91)
... Ind[Nov 2010–Dec 2010]		-0.00606 (0.056)		-0.00602 (0.0048)
... Ind[Jan 2011–Feb 2011]		-0.0103* (0.0062)		-0.00732 (0.0050)
... Ind[Mar 2011–Apr 2011]		-0.00247 (0.0070)		0.00307 (0.0077)
Trader FE	x	x	x	x
Day FE	x	x	x	x
Bid-ask quotes matched perfectly to trades	Yes	Yes	No	No
Controls for match distance in log(seconds)	No	No	Yes	Yes
Number of trades	920,573	920,573	1,444,159	1,444,159
R ²	0.016	0.016	0.015	0.015

Panel B: The January 2010 regulation announcement and relative bid-ask spreads				
Dep var: Relative bid-ask spreads	(1b)	(2b)	(3b)	(4b)
US trader (=1) × Post announcement (=1)	-0.00940 (0.013)	0.00789 (6.16)	-0.00586 (0.0099)	0.0110 (0.33)
US trader (=1) × ...				
... Ind[Nov 2009–Oct 2009]		-0.0118 (5.96)		-0.00869 (2.42)
... Ind[Nov 2009–Dec 2009]		-0.0141 (1.67)		-0.0105 (1.01)
... Ind[Jan 2010–Mar 2010]		-0.0235 (0.015)		-0.0200 (0.013)
... Ind[Apr 2010–May 2009]		-0.0227 (0.019)		-0.0217 (0.017)
... Ind[Jun 2009–Jul 2009]		-0.0186 (0.040)		-0.0135 (0.036)
Trader FE	x	x	x	x
Day FE	x	x	x	x
Bid-ask quotes matched perfectly to trades	Yes	Yes	No	No
Controls for match distance in log(seconds)	No	No	Yes	Yes
Number of trades	644,755	644,755	1,140,274	1,140,274
R ²	0.017	0.018	0.013	0.013

Read:

- The average relative bid-ask spread is close to zero and even slightly negative on average, consistent with brokers often filling from their own inventory.
- The difference-in-differences coefficient on *US × PostConstraint* is about **-0.0105** in exact-second matches and about **-0.0138** in the broader matched sample.
- Neither estimate is statistically distinguishable from zero.
- The announcement-period spread regressions are also insignificant.

Economic message. This is the punch line for market quality. The leverage cap reduced volume and intermediary rents, but there is **no sign** that it made execution prices worse for the remaining traders.

6 Short summary

- The 2010 U.S. retail-FX leverage cap creates a sharp policy experiment.
- Relative to comparable European traders, U.S. traders cut trading volume by about **23%**.
- The policy especially helps the most levered traders, whose large pre-policy losses shrink dramatically.

- Brokers' excess capital falls by about **25%**, especially among small and previously high-leverage brokers.
- Relative bid-ask spreads do **not** rise.
- The evidence supports the view that leverage in this market mainly financed **costly speculation**, not socially useful liquidity provision.

7 Conclusion

7.1 Core conclusion

The paper's bottom line is clear: in the U.S. retail foreign-exchange market, restricting leverage substantially reduces trading activity and trader losses, while shrinking the brokerage sector, *without* producing a detectable liquidity cost in the form of wider spreads.

7.2 Economic intuition

The most compelling interpretation is that retail leverage was helping traders take oversized speculative positions that generated large round-trip volume and large spread payments. Those trades enriched the intermediaries that facilitated them, but did not create enough informational or liquidity value to justify the resource cost of intermediation.

7.3 Takeaway

If this market is representative of at least some other segments of retail finance, then leverage can be a policy target for reducing socially excessive speculative intermediation. The striking feature of this paper is not only that the leverage cap helps traders; it is that it seems to do so **without** harming market quality on the main liquidity margin the authors can observe.